

Bridging Discrepancies in Debate: Joint Debate for Enhancing Multimodal Emotion-Cause Pair Extraction in Conversation

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Abstract—Multimodal Emotion-Cause Pair Extraction in Conversation (MECPEC) aims to accurately identify emotions and their causes in multimodal multiparty conversations. Existing methods primarily rely on discriminative or generative predictions from pre-trained language models, overlooking the potential to leverage the differences and complementary strengths between these two approaches. This paper proposes a novel framework, DEC-Debate, based on a multiparty-multiround debate mechanism. Concretely, the utterance emotion classification results from different models are integrated through debate, which can assign more appropriate emotion labels to utterances. Subsequently, multiple LLMs collaborate to combine their knowledge, express the causal relationships in each emotion-cause utterance pair, and then explain and update them through multi-round debate, reducing the impact of erroneous model knowledge. By enabling independent generation and mutual criticism, our method gradually reaches consensus, enhancing the accuracy and reliability of the inference. It achieves state-of-the-art performance and strong generalization on three public benchmark datasets.

Index Terms—Affective computing, emotion analysis, emotion-cause pair extraction, LLMs debate

I. INTRODUCTION

WITH the rapid development of social media, emotion-cause analysis has shown great potential in areas such as public opinion analysis, mental health, and intelligent recommendation, and has thus become a research hotspot in affective computing. Lee et al. [1] first proposed emotion-cause extraction (ECE), a foundational task of emotion cause analysis. To address the issue that ECE requires additional emotion annotations, Xia and Ding [2] later extended the ECE to the more comprehensive emotion-cause pair extraction

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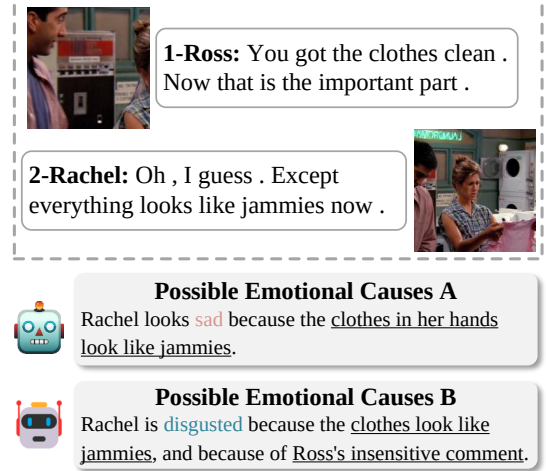


Fig. 1. An illustration of the challenges in emotion classification and emotion-cause pair reasoning in real-world conversations. The example provides two perspectives: one infers Rachel’s emotion as sadness based on facial expressions, but overlooks the impact of Ross’s insensitive comment on this emotion; the other captures more completely the causal clues leading to Rachel’s emotion, but relies more heavily on these clues to infer the specific emotion category.

(ECPE) task [3, 4]. However, these document-level methods cannot be applied to the large number of real-time communication scenarios prevalent in modern applications. Some methods have begun to focus on conversational emotion-cause analysis [5, 6, 7]. Meanwhile, relying solely on textual information fails to effectively capture the multimodal cues, such as audio and visual signals, that exist in the real world. Consequently, multimodal emotion recognition has attracted increasing attention [8, 9, 10, 11]. To fill the gap in multimodal emotion-cause pair extraction, Wang et al. [12] proposed the MECPEC task, which aims to extract emotion-cause pairs from multimodal multiparty conversations and has broad application prospects.

By effectively addressing complex and polysemous emotional expressions, MECPEC contributes to enhancing the empathy capability of human-computer interaction (HCI) and promotes its shift from instruction-based interaction to proactive interaction. For instance, in mental health chatbots, identifying the precise cause of a user’s emotion allows the system to generate compassionate and relevant responses rather than generic words of comfort. In customer service systems, recognizing that anger stems from ‘late delivery’ rather than ‘prod-

uct quality’ helps route the issue to the appropriate department and tailor resolution strategies. Thus, MECPEC has generated significant interest among researchers and established it as a highly notable point in the current affective computing field.

Existing research focuses mainly on discriminative modeling [13, 14] or generative modeling [15, 16, 17]. Even though task performance has improved, certain limitations remain: 1) Discriminative models tend to rely on superficial utterance cues, leading to shortcut learning [18] and capturing spurious emotion-cause correlations instead of genuine causal structures; 2) Generative models may suffer from hallucinations [19], causing over-interpretation of the context and fabricating non-existent or incorrect emotion-cause pairs. However, these two approaches exhibit a certain degree of complementarity: discriminative methods can help constrain the excessive emotion cause reasoning in generative methods, while generative methods can provide richer implicit emotion cause predictions. To leverage these complementarities to integrate the advantages of both approaches, we propose a dual model prediction discrepancies-based emotion cause debate (DEC-Debate) framework, which establishes a multi-perspective reasoning mechanism with mutual rebuttals to uncover more plausible emotion classifications and emotion-cause pairs from the prediction discrepancies between different models.

As shown in Fig. 1, reasoning about emotion-cause from different perspectives can yield distinct but reasonable inferences about the same utterance, reflecting subjectivity and context-dependency. Differences in model architecture and learning mechanisms cause each model to develop a unique way of understanding utterances and context. As a result, different models may produce divergent predictions, yet each prediction is consistent with its internal reasoning. Namely, these discrepancies are not noise and can highlight uncertain or challenging samples. Therefore, the discrepancies among different models’ emotion-cause predictions should not be discarded, but leveraged as starting points for further reasoning to classify emotions and uncover potentially valid emotion-cause pairs within them. Interestingly, large language models (LLMs) exhibit preliminary human-like reasoning capabilities and can generate explanations for emotions and emotion-cause pairs, offering a promising tool for such further reasoning. Nevertheless, LLM outputs may contain errors or fabrications, making the reliance on a single LLM suboptimal. To this end, it is necessary to establish an effective and structured mechanism. Debate [20] is a structured process in which all parties orderly elaborate their respective views and consider opposing views, reaching a more convincing and reasonable decision through multiparty reasoning [21]. And some research indicates that, in generative tasks, the use of multi-LLMs can help improve output quality [22, 23]. Furthermore, debate can be modeled as a non-cooperative game [24]. Since MECPEC is the process of abductive reasoning, this paper introduces a debate mechanism guided by non-cooperative game theory with two modules: emotional boundary argument (EBA) and emotion-cause pair retention argument (ECPRA). Specifically, discrepancies among the predictions of different models will trigger debate among preconfigured LLMs: each LLM first

selects the prediction it deems more plausible and provides an explanation, then evaluates the responses of others to update its own judgment, and adjusts both its choice and explanation in the next round. EBA encourages LLMs to contextualize emotional scenes and argue with each other to determine the optimal emotion category for a given utterance, refining emotion classification boundaries; ECPRA promotes LLMs to reason through clues by evaluating whether the event leading to the target emotion exists in the causal utterance or its potential background, thus eliminating the unreasonable pairs and only retaining the pairs with clear supporting evidence. Finally, majority voting ensures the objectivity of the debate results. The primary contributions of this paper are summarized as follows:

- The debate mechanism is introduced into the tasks of conversational emotion analysis and MECPEC for the first time.
- We propose a multiparty-multiround debate framework with two core modules: EBA and ECPRA, improving the accuracy of the emotion category classification and the extraction of emotion-cause pairs respectively.
- The experiment results demonstrate that DEC-Debate achieves state-of-the-art (SOTA) results on the ECF for the MECPEC task and shows excellent generalization capabilities on the MELD and ConvECE.

II. RELATED WORK

Current methods of emotion-cause analysis can be roughly categorized into discriminative and generative methods. The abductive reasoning mechanisms of these two methods are different. Discriminative methods identify emotions and their causes by modeling characteristic correlations between utterances. In contrast, generative methods tend to directly produce human-like descriptive outputs of the relationships between emotions and their causes.

A. Discriminative methods for Emotion-Cause Analysis

In ECE, since emotion words are given annotations, some work [25, 26] enable the model to learn to identify keywords within cause clauses and the association patterns between these keywords and the target emotion, thereby determining whether a candidate clause is a cause clause through binary classification. However, Hu et al. [27] point out that previous studies have neglected document-level context, and proposed combining a bidirectional hierarchical attention network with a sentiment filtering mechanism to both utilize document-level contextual information and filter out noise. For ECPE, typical methods [28, 29, 30] involve interacting and fusing the embeddings of candidate emotional and causal clauses to generate a joint representation, based on which binary classification is performed to predict whether the pair constitutes an emotion-cause relationship. Hu et al. [31] introduce a contrastive learning mechanism to pull positive sample pairs closer together and push negative sample pairs farther apart. Notably, Gu et al. [32] adopt a prompt-template-based end-to-end approach to transform emotional clause extraction, causal clause extraction, and pair prediction into a masked language

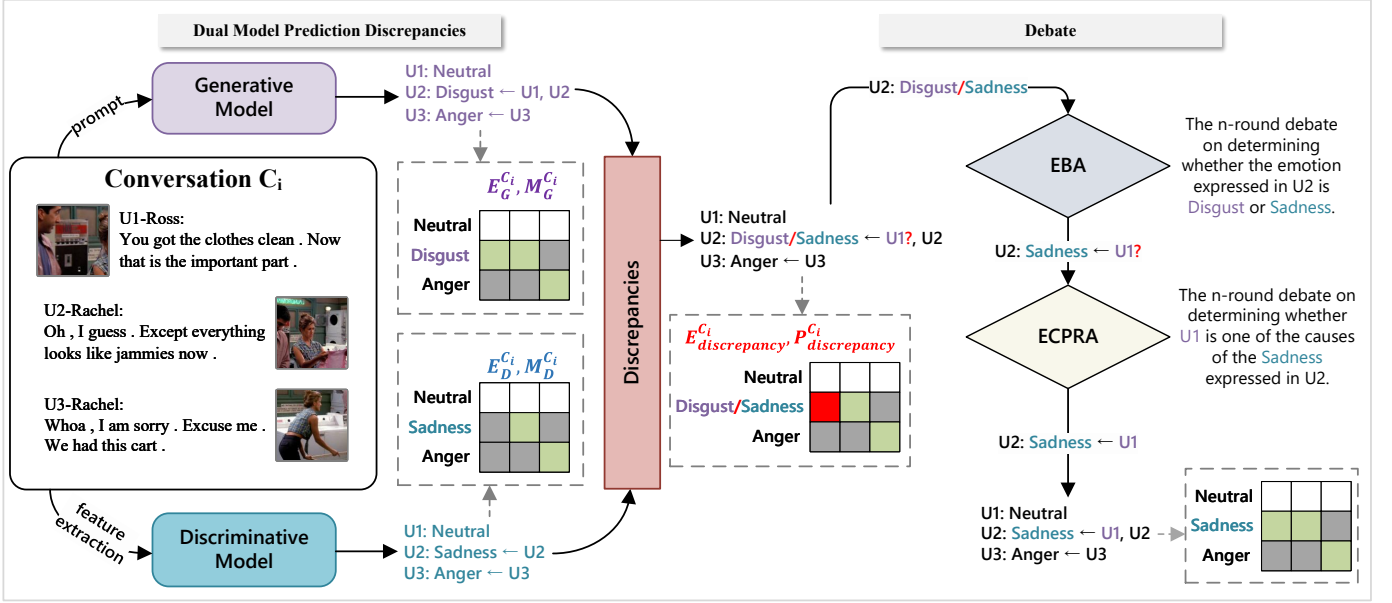


Fig. 2. Divergent in emotion classification for the same utterance and in emotion-cause pairs within the conversation, as predicted by two models, are extracted and processed by EBA and ECPRA for decision-making, respectively. Emotion-cause pair distributions in the conversation are represented as matrices, with rows for emotional utterances and columns for causal ones. Gray, green, and red cells denote candidate pairs, predicted pairs, and conflicting pairs, respectively.

modeling task, thus changing this paradigm. Conversational settings impose higher demands on ECPE to understand fragmented utterances and capture information flow. Wang et al. [33] construct a conversational event propagation mechanism that enables the representation of emotional utterances to incorporate the contextual information of their corresponding event, thus enhancing the association between emotional and causal utterances and improving the pairing probability. However, the discriminative method based on joint representation is essentially a black-box association judgment. While it can determine whether a pair of sentences or utterances constitutes a specific relation, it lacks the ability to generate readable reasoning chains or formal proofs.

B. Generative methods for Emotion-Cause Analysis

Given the stronger semantic understanding and natural language generation capabilities of LLMs such as ChatGPT [34], generative approaches built around LLMs are beginning to transform emotion cause classification into generative prediction. These methods typically employ carefully designed natural language prompts to instruct the LLM to directly generate the emotion category of an utterance or sentence, along with the position of its corresponding cause. Some works [35, 36] employ a basic approach to extract emotion cause, first by asking the LLM "What emotion is expressed in this utterance?" and then, based on the emotion category provided by the LLM, further querying the LLM, "What is the cause of this emotion?". This approach is simple and intuitive, but the output is fragile and its accuracy heavily depends on the performance of the LLM used. Huang et al. [37] mimic the cognitive process of human emotion generation, guiding LLMs to first summarize the overall theme of the conversation, then identify the specific reactions expressed in the utterances,

next evaluate the speaker's internal psychological state that gives rise to these reactions, and finally trace back the external triggering factors that lead to this state, thereby accomplishing a step-by-step reasoning for identifying the emotion cause. Rusnachenko and Liang [38] also adopted a similar chain-of-thought approach. Although the emotion-cause reasoning process has been refined and deepened, its effectiveness has been questioned [39]. In summary, generative approaches hold greater potential due to their ability to provide human-like descriptions of the reasoning process, but they also carry the inherent risk of generating incorrect content.

III. METHODOLOGY

A. Task Definition

In a multiturn conversation $C = \{U_1, U_2, \dots, U_{|C|}\}$, each utterance contains three modalities: text, video, and audio, i.e., $U = \{t, v, a\}$. The goal of the MECPEC task is to identify the emotion in each utterance of a conversation and, within the same conversation, to find the causal utterances for each emotional utterance. The MECPEC task consists of two subtasks: emotion recognition and emotion-cause pair extraction. The former aims to combine multimodal information to classify each utterance as either emotional or causal (e.g., $\hat{y}^e = F_e(t, v, a)$, $\hat{y}^c = F_c(t, v, a)$), with emotional utterance further categorized into Ekman's six classifications of emotions [40] ("anger", "disgust", "fear", "joy", "sadness" and "surprise"). However, understanding the causes behind emotions is even more important. The latter further identify the causes of utterances emotion within the same conversation. Its typical approach is first form candidate pairs between the emotional utterances and the causal utterances, then predict the correct pairs and extract them.

B. Overall Framework

As shown in Fig. 2, the DEC-Debate consists mainly of two steps. We first constructed two models, one based on the generative model (Fig. 3) and the other on the discriminative model (Fig. 4), to perform utterance-level emotion recognition and emotion-cause pair extraction in the target conversation, respectively. The predictions from the two models are compared to identify areas of agreement and disagreement. Specifically, they disagree on both the emotion category of U2 (Rachel) and whether U1 (Ross's comment) constitutes a cause of Rachel's emotion in U2. These discrepancies are then fed into the subsequent debate mechanism.

The entire debate process consists of two modules: Emotional Boundary Argument (EBA) and Emotion-Cause Pairs Retention Argument (ECPRA). In the EBA phase, multiple LLMs act as debaters and engage in n rounds of discussion, leveraging contextual information to determine the emotion category of U2. A consensus is finally reached through majority voting, concluding that Rachel's emotion is "Sadness." Based on this outcome, the ECPRA module is activated to conduct another multiple LLMs debate on whether U1 (Ross's comment) is one of the causes of Rachel's emotion. After n rounds of interaction, the final decision confirms that U1 is indeed a contributing cause of Rachel's emotion in U2.

C. Dual Model Prediction Discrepancies

We have constructed two predictive models: an LLM-based generative model (Fig. 3) and a BERT [41]-based discriminative model (Fig. 4), which demonstrate distinct and complementary strengths in emotion-cause analysis. The generative model, built upon LLM, leverages the LLMs' internalized commonsense knowledge and excellent contextual understanding, excelling at identifying implicit emotions and causes. However, due to its autoregressive nature, it tends to generate over-interpretations that go beyond the available evidence, reflecting a bias toward excessive inference. In contrast, the discriminative model reliably detects explicit emotional causes by learning direct associations between utterance features and emotion-cause labels, often misses subtle or highly context-dependent cues, indicating a conservative, pattern-driven bias. As a result, the two models are more likely to produce prediction discrepancies on challenging samples. These discrepancies refer to divergent emotion categories of the same utterance provided by different models, or to the symmetric difference set of emotion-cause pairs extracted from the same conversation by different models. Simply combining the predicted results of the two models will propagate rather than mitigate existing errors, degrading overall performance. Instead, these discrepancies highlight opportunities for improvement. By identifying more plausible predictions in them, we can reduce individual model biases and better leverage their complementary strengths, making more robust decisions.

Specifically, given a conversation C_i consisting of $|C|$ utterances, the two different models generate their respective predictions. In the emotion recognition subtask, the generative model and the discriminative model separately provide

pseudo-label sets for the emotional classification of each utterance in the conversation, denote as $E_G^{C_i}$ and $E_D^{C_i}$ respectively:

$$E_G^{C_i} = \{\hat{e}_G^1, \hat{e}_G^2, \hat{e}_G^3, \dots, \hat{e}_G^{|C|}\} \quad (1)$$

$$E_D^{C_i} = \{\hat{e}_D^1, \hat{e}_D^2, \hat{e}_D^3, \dots, \hat{e}_D^{|C|}\} \quad (2)$$

The emotion pseudo-labels provided by the generative model ($\hat{e}_G^j$) and the discriminative model ($\hat{e}_D^j$) are fused to reconstruct a new pseudo-label set E^{C_i} :

$$E^{C_i} = \{(\hat{e}_G^1, \hat{e}_D^1), \dots, (\hat{e}_G^{|C|}, \hat{e}_D^{|C|})\} \quad (3)$$

All elements in set E^{C_i} where $\hat{e}_G^j \neq \hat{e}_D^j, j \in [1, |C|]$ are extracted to form the emotion prediction discrepancy set $E_{discrepancy}^{C_i}$.

In C_i , the generative model predicts emotion-cause pairs and constructs an utterance pairing state matrix $M_G^{C_i} = \{(m_G^{jk} \in \{0, 1\}) | j, k \in [1, |C|]\}$. When $m_G^{jk} = 1$, U_j is a non-neutral utterance and U_k contains its emotional cause. If $m_G^{jk} = 0$, U_j and U_k do not form a pair. Similarly, the discriminative model's predictions yield another pairing state matrix $M_D^{C_i} = \{(m_D^{jk} \in \{0, 1\}) | j, k \in [1, |C|]\}$. Extract the non-overlapping regions from the two matrices to form the set $P_{discrepancy}^{C_i}$ of non-overlapping pairs. The overlapping pairs form the set $P_{common}^{C_i}$.

$$P_{discrepancy}^{C_i} = \{(U_j, U_k) | m_G^{jk} \neq m_D^{jk}\} \quad (4)$$

$$P_{common}^{C_i} = \{(U_j, U_k) | m_G^{jk} = m_D^{jk} = 1\} \quad (5)$$

The process is summarized in Algorithm 1.

Algorithm 1: Dual Model Prediction Discrepancies

Input: conversation C_i ;
Output: $E_{discrepancy}^{C_i}, P_{discrepancy}^{C_i}$;

- 1 model1 = Generative Model ;
- 2 model2 = Discriminative Model ;
- 3 $|C| = \text{len}(C_i)$;
- 4 $E_{discrepancy}^{C_i} = []$;
- 5 $P_{discrepancy}^{C_i} = []$;
- 6 $E_G^{C_i}, M_G^{C_i} = \text{model1}(C_i)$;
- 7 $E_D^{C_i}, M_D^{C_i} = \text{model2}(C_i)$;
- 8 **for** $j = 1; j \leq |C|$ **do**
- 9 $e_G^j = E_G^{C_i}[j]$;
- 10 $e_D^j = E_D^{C_i}[j]$;
- 11 **if** $e_G^j \neq e_D^j$ **then**
- 12 $E_{discrepancy}^{C_i}.\text{append}((e_G^j, e_D^j))$;
- 13 $P_{discrepancy}^{C_i} = \text{non-overlapping}(M_G^{C_i}, M_D^{C_i})$;
- 14 **return** $E_{discrepancy}^{C_i}, P_{discrepancy}^{C_i}$;

D. Debate

The design of the debate mechanism adapts and draws on concepts of game theory, and can be modeled as a stylized non-cooperative game. This game-theoretic debate involves m LLMs acting as debaters, forming a player set $\mathcal{M} = \{1, 2, \dots, m\}$. In each round of debate, all debaters jointly

address a target x (e.g., determining the emotion classification of a given utterance, or deciding whether a specific emotion-cause pair should be retained). Strategy π_i of debater i is to independently output its chosen option y_i^t for the t -th round and the natural language rationale r_i^t supporting this decision, based on the input x and the public action history h^{t-1} . In our approach, π_i incorporates an objective of refuting positions considered unreasonable, thereby enabling flawed emotion causality to be identified and weakened through iterative rebuttals.

$$\pi_i : x, h^{t-1} \rightarrow (y_i^t, r_i^t), y \in \mathcal{Y}, r \in \mathcal{R} \quad (6)$$

$$h^t = \{(y_i^t, r_i^t)\}_{i=1}^m, h^0 = \emptyset \quad (7)$$

$$\mathcal{Y} = \{\text{optionA}, \text{optionB}\}, \mathcal{R} = \text{Natural Language} \quad (8)$$

This process embodies multi-round interactive reasoning, where each debater can independently update its stance by drawing on the previous decisions and arguments of other debaters. The game is deemed to approach an equilibrium state when further rounds j of debate do not significantly improve overall performance ($Perf$). The fixed value n of rounds that approaches this equilibrium state can be obtained through multiple experiments (i.e., $t : 1 \rightarrow n$). At this point, the debate concludes, and the decision that garners the most support in the final round is adopted as the outcome.

$$Perf(\{y_i^n\}_{i=1}^m) \approx \frac{\sum_{t=n}^{n+j} Perf(\{y_i^t\}_{i=1}^m)}{j+1} \quad (9)$$

Two main modules, Emotional Boundary Argument (EBA) and Emotion-Cause Pairs Retention Argument (ECPR), are constructed based on the multiple LLMs debate mechanism to process $E_{discrepancy}^{C_i}$ and $P_{discrepancy}^{C_i}$, respectively. EBA focuses on selecting a more appropriate emotion classification between $\hat{e}_G$ and $\hat{e}_D$, while ECPR is responsible for choosing more reasonable emotion-cause pairs from $P_{discrepancy}^{C_i}$. The process is summarized in Algorithm 2.

Algorithm 2: Debate

Input: $E_{discrepancy}^{C_i}$, $P_{discrepancy}^{C_i}$, $P_{common}^{C_i}$;
1 m debaters, n debate rounds ;
2 **for** each e in $E_{discrepancy}^{C_i}$ **do**
3 buffer $\leftarrow$ EBA(e , m) ;
4 **for** $l = 2; l \leq n$ **do**
5 buffer $\leftarrow$ EBA(e , m, buffer) ;
6 utterance $\hat{e} \leftarrow$ major emo in buffer ;
7 retained pairs = [] ;
8 **for** each p in $P_{discrepancy}^{C_i}$ **do**
9 buffer $\leftarrow$ ECPR(p , m) ;
10 **for** $l = 2; l \leq n$ **do**
11 $\hat{e} =$ emotion of p ;
12 buffer $\leftarrow$ ECPR(p , $\hat{e}$, m, buffer) ;
13 **if** major result in buffer is "yes" **then**
14 append(retained pairs, p) ;
15 extend($P_{common}^{C_i}$, retained pairs) ;

1) *Emotional Boundary Argument:* For the $E_{discrepancy}^{C_i}$, the following topic is generated for each element: "In this conversation $\langle C_i \rangle$, which is the most likely emotion expressed in $U_j, j \in [1, |C|]$? (A) $\langle \hat{e}_G^j \rangle$ (B) $\langle \hat{e}_D^j \rangle$. Choose only one option and provide a brief explanation". These topics are addressed sequentially through debate. One topic is fed into m LLMs (debaters) participating in the debate to obtain the results of the first round of debate, and each debater's choice and explanation are stored in a buffer as results. For n rounds of debate, if the current round number is l and $l > 1$, the results of the $(l-1)$ -th round stored in the buffer are added to the topic for the l -th round (e.g., " $\langle (l-1)$ -th round results $\rangle$, you are debaterX. In your view, are the opinions of the other debaters more reasonable or less reasonable?"), which is then fed into each debater. Meanwhile, the new results are updated to the buffer.

Based on the results of the final round of debate, the final emotional classification $\hat{e}^j$ of U_j in conversation C_i is corrected to the one preferred by the majority of debaters between $\hat{e}_G^j$ and $\hat{e}_D^j$.

2) *Emotion-Cause Pairs Retention Argument:* Each element of the set $P_{discrepancy}^{C_i}$ is used to construct the following topic: "In this conversation $\langle C_i \rangle$, U_j 's emotion is $\langle \hat{e}^j \rangle$. Is there a reason in U_k that leads to the emotion in U_j ? (A) yes (B) no. Choose only one option and provide a brief explanation" ($j, k \in [1, |C|]$). The topic is then fed into m debaters for n rounds of debate.

Based on the results of the final round, if the majority of debaters agree that, in conversation C_i , utterance U_k contains the cause of the emotion in utterance U_j , the pair (U_j, U_k) is retained; otherwise, it is discarded. Retained pairs are added to $P_{common}^{C_i}$ as the final set of emotion-cause pairs for C_i .

E. Generative Model Predicting

Generally, LLM-based generative methods employ simple zero-shot question-answer generation approaches to extract utterance emotion and causes. However, such approaches often produce uncontrollable outputs and fail to yield satisfactory performance. Therefore, inspired by fine-tuning methods for pre-trained models, we design a template to fine-tune the LLM. Specifically, we construct input prompts with masked slots for emotional and causal labels (e.g., [CLASS], [IDS]), and train the LLM to fill in these slots. For example (Fig. 3), a prompt like "'Oh, I guess. Except everything looks like jammies now.'" contains [CLASS] emotion, corresponding to [IDS]." Here, [CLASS] is the emotion classification mask of the utterance, [IDS] is the list of utterance numbers where emotional causes are located. These prompts enable LLM to generate emotional classification pseudo labels ($E_G^{C_i}$) for utterances and identify the positions of their corresponding causal utterances ($M_G^{C_i}$) within the given conversation C_i in an end-to-end manner.

F. Discriminative Model Predicting

To validate the effectiveness of the debate mechanism in a controlled and reproducible setting, we employ a standard BERT [41]-based architecture (Fig. 4) without elaborate optimizations or complex designs to build the discriminative

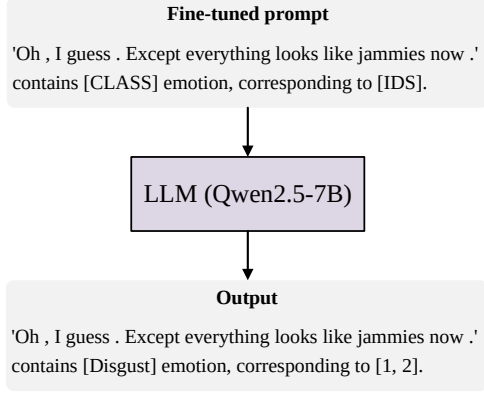


Fig. 3. Based on the performance comparison, we have decided to fine-tune a Qwen2.5-7B to build the generative model predicting.

model. Utterance features are extracted using BERT [41] (text), openSMILE [42] (audio), and 3D-CNN [43] (video), respectively. The features of the three modalities are first uniformly reduced to 768 dimensions. Afterward, the multimodal features are fused with an attention-based multimodal shifting fusion method[44].

$$Z^{768} = \text{Fusion}(F_{\text{text}}^{768}, F_{\text{video}}^{768}, F_{\text{audio}}^{768}) \quad (10)$$

The fused features are fed into two separate multilayer perceptron (MLP) to identify emotional and causal utterance. Features of the emotional and causal utterance are concatenated and fed into BiLSTM [45] to determine whether there is an emotion-cause relationship between the two utterances.

$$\hat{y}_D^{e7} = \text{MLP}_e(Z^{768}), \hat{y}_D^{c2} = \text{MLP}_c(Z^{768}) \quad (11)$$

$$\hat{y}_D^{p2} = \text{MLP}_{\text{pair}}(\text{BiLSTM}([Z^{e768}, Z^{c768}])) \quad (12)$$

$$\hat{e}_D = \text{argmax}(\hat{y}^{e7}), m_D^{ec} = \text{argmax}(\hat{y}_D^{p2}) \quad (13)$$

The superscript numbers on the variables indicate their feature dimensions. Considering the identification of "neutral" utterance, the emotion features are set to 7 dimensions.

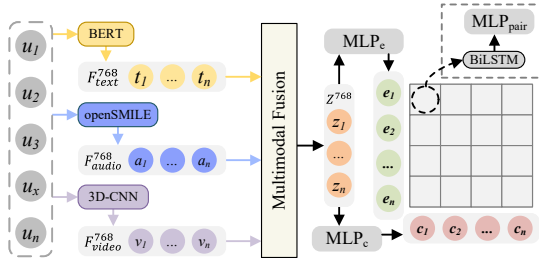


Fig. 4. A generic discriminative model for MECPEC.

G. Modality Translation

Due to the current scarcity of mature, fully multimodal generative LLMs, this study did not employ such models for foundational generative prediction nor as debaters in EBA and ECPRA. However, we leverage a fully multimodal LLM

to recognize visual and acoustic modalities and translate them into text, which is then integrated into the conversation samples. These enhanced samples were subsequently used for generative model predicting and debate, thereby further endowing the model with multimodal perception capabilities. In our experiments, we use the current SOTA model, Qwen2.5-Omni-3B [46]. As shown in Fig. 5, textual descriptions of the speaker's behavior and tone are incorporated into the conversation utterances.

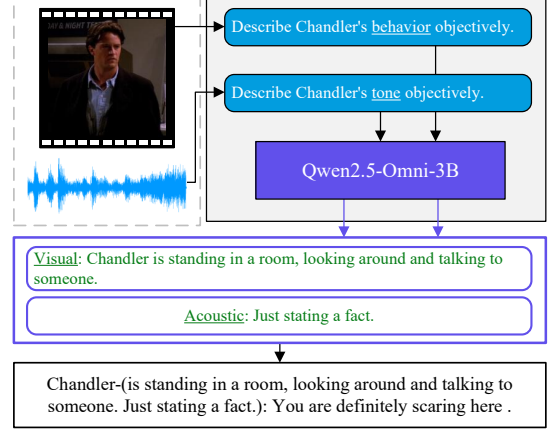


Fig. 5. The process by which a multimodal LLM converts visual and acoustic information into text that can be ingested by text-based generative LLMs.

TABLE I
A SUMMARY OF THE ECF, MELD, AND CONVECEPE DATASETS.

	ECF	MELD	ConvECPE
Modality	T, A, V	T, A, V	T, A, V
Conversations	1,374	1,433	151
Utterances	13,619	13,708	7,433
Emotions	7,690	5,433	5,725
Pairs	9,794	0	9,473

IV. EXPERIMENTS

A. Datasets

The performance of our approach is evaluated on the ECF [12] dataset and its generalizability is verified on the MELD [47] and ConvECPE [48] datasets. An overview of these datasets is presented in TABLE I. The ECF contains 1,374 conversations and 13,619 utterances, and each utterance containing information from three modalities: text, video, and audio. MELD includes 1,433 conversations and only labels utterance emotions without annotating corresponding causes. Therefore, for MELD, only the generalization performance of emotion recognition is validated. ConvECPE differs significantly from ECF. Firstly, ConvECPE employs five emotion labels: happiness, sadness, anger, excitement and frustration. Secondly, while ECF consists of multiparty and multi-turn conversations, ConvECPE contains dualparty, multi-turn conversations with a larger number of conversations turns. However, ConvECPE also annotates emotional cause. Therefore, the generalization performance of emotion-cause pair extraction can be validated on ConvECPE.

B. Settings

LoRA [49] and the designed prompt III-E are applied to fine-tune (rank, dropout, epochs, learning rate = 8, 0, 6, 5e-5) three open-source LLMs: Qwen2.5-7B [50], GLM4-9B [51], and Llama3.1-8B [52]. The fine-tuned Qwen2.5-7B performs better with fewer nonsense and is applied as the generative model for predicting. The discriminative model is trained for 5 epochs with a batch size of 2 in emotion recognition, and it is trained for 60 epochs with a batch size of 1 in emotion-cause pair extraction. Both stages use the Adam optimizer with a learning rate of 1e-5. For the multi-LLM debate, we selected several open-source quantized LLMs with comparable performance¹ (TABLE II) for experiments ($m=5$). Ultimately, Qwen2.5-7B, GLM4-9B, and Llama3.1-8B are chosen as debaters ($m=3$), as they demonstrate strong and similar performance on key capabilities including reasoning and language understanding, thereby achieving the best overall results. The temperature of these models was set to 0.7. To avoid unduly constraining the reasoning processes of the debaters involved in the n rounds debate ($n=2$), fine-tuning is not performed on them. All experiments are conducted on a platform equipped with NVIDIA GeForce RTX 4090 GPUs.

TABLE II
BENCHMARK SCORES. OPEN-SOURCE AND QUANTIZED LLMs SUFFICE FOR OUR EXPERIMENTS WHILE LOWERING COMPUTATIONAL COSTS.

Models	Reasoning	Language	Scale
Qwen2.5 [50]	46.4/100	45.8/100	7B
GLM4 [51]	44.6/100	45.5/100	9B
Llama3.1 [52]	45.0/100	45.7/100	8B
InternLM2.5 [53]	39.2/100	42.1/100	7B
Gemma2 [54]	34.4/100	45.0/100	9B

In the main experiments, the evaluation metrics employed are Precision (P), Recall (R), F1, and weighted-average F1 ($w\text{-avg. F1}$), which are computed as follows:

$$P = \frac{\text{correct}}{\text{predicted}}, P^{\text{cate}} = \frac{\sum \text{correct}^{\text{cate}}}{\sum \text{predicted}^{\text{cate}}} \quad (14)$$

$$R = \frac{\text{correct}}{\text{annotated}}, R^{\text{cate}} = \frac{\text{correct}^{\text{cate}}}{\text{annotated}^{\text{cate}}} \quad (15)$$

$$F1 = \frac{2PR}{P+R}, F1^{\text{cate}} = \frac{2P^{\text{cate}}R^{\text{cate}}}{P^{\text{cate}}+R^{\text{cate}}} \quad (16)$$

$$w^{\text{cate}} = \frac{\text{annotated}^{\text{cate}}}{\text{annotated}^{\text{all-cates}}} \quad (17)$$

$$w\text{-avg. F1} = \sum_{\text{cate}} w^{\text{cate}} F1^{\text{cate}} \quad (18)$$

Where *correct*, *predicted*, and *annotated* denote the number of correctly predicted samples, the number of predicted samples, and the number of annotated samples, respectively. Their definitions vary by task: in emotion-cause pair extraction, a sample is an emotion-cause utterance pair; in emotion recognition, it is an individual utterance. *cate* refers to an emotion category, and w is its proportion (weight) in the dataset. $w\text{-avg. F1}$ is the weighted sum of $F1$ scores across

all categories. $w\text{-avg.6}$ excludes "Neutral"; $w\text{-avg.4}$ excludes "Neutral," "Disgust," and "Fear."

C. Baselines

The baseline methods we selected can be roughly categorized into two types: discriminative methods and LLM-based generative methods.

The former mainly focuses on how to find clues of emotional causes in feature space. For example, in previous SOTA methods: UniMEEC [55] used mask prediction to discover causal relationships between emotional and causal utterances; HiLo [56] enhances feature interaction by introducing masks for three distinct perspectives; M³HG [57] models emotional causal dependencies in conversations by optimizing node information within a multimodal graph structure; AHMRC [58] introduces an adaptive multimodal fusion mechanism and transforms the emotion-cause pair prediction into a two-round query task. Additionally, some methods try to enhance features by leveraging additional information, such as improving multimodal feature fusion (MECPEC-2steps [12], UniMSE [59] and GA2MIF [60]) and context information integration (MMGCN [61], MM-DFN [62] and TSAM [63]).

For the latter, GMEC [64] employs a discriminative model to perform preliminary reasoning on multimodal conversations, generating candidate answers and their confidence scores, which are then incorporated into prompts for a generative LLM to make the enhanced decision. However, it does not establish a deeper collaborative relationship between the LLM and the discriminative model. Apart from GMEC, methods that directly employ LLMs for MECPEC remain relatively few, and their performance has generally not shown significant advantages. Therefore, for completeness, we constructed baselines using vision-LLMs: LLaVA, Video-Llama, and Qwen2.5-Omni. Specifically, we applied the same prediction generation method mentioned in III-E, extending it to accommodate additional visual and video-audio modalities to fine-tune them.

D. Main Results

The main results of emotion-cause pair extraction are shown in TABLE III. Our approach outperforms SOTA in most key metrics. In the $w\text{-avg.6}$ metric, it surpasses the previous SOTA models (HiLo [56], M³HG [57], and GMEC [64]) by 12.59%, 5.56%, and 4.11% respectively. In the F1 score for emotion-cause pair extraction, it outperforms the previous SOTA models (UniMEEC [55], HiLo [56], AHMRC [58], and GMEC [64]) by 7.12%, 6.28%, 4.13%, and 2.28% respectively. These improvements can be summarized as: 1) Our approach progressively refines emotion-cause pair extraction through debate and further enhances the rationality in the predictions, establishing a clear causal relationship for emotion-cause pair; 2) Debate utilizes the advantages of a generative model and a discriminative model. Overall, DEC-Debate extracts the more accurate components from the discrepancies, thereby enhancing the overall prediction accuracy, which is evidenced by the increase in the precision. Meanwhile, the performance of emotion recognition (ER) and cause extraction (CE) is also shown in TABLE III. ER is

¹<https://rank.opencompass.org.cn/leaderboard-llm>

TABLE III

RESULTS ON ECF TEST DATASET. BOLD INDICATES THE OPTIMAL RESULT, AND UNDERLINE INDICATES THE SUBOPTIMAL RESULT. THE BASELINES WITH * ARE REPRODUCED USING THEIR OPEN-SOURCE CODE BY HU ET AL. [55].

Methods		P	Emotion-Cause Pair		Extraction		ER	CE
			R	F1	w-avg.6	w-avg.4	w-avg.F1	F1
Discriminative Models	MECPEC-2steps [12]	0.5727	0.5216	0.5451	0.3062	0.3343	-	0.6987
	TSAM [63]	-	-	0.3870	0.3774	-	-	-
	MMCGN* [61]	0.3543	0.3819	0.3748	-	-	0.5465	0.5530
	MM-DFN* [62]	0.3790	0.3908	0.3810	-	-	0.5486	0.5517
	UniMSE* [59]	0.4448	0.5425	0.4908	-	-	0.5637	0.5673
	GA2MIF* [60]	0.4615	0.5426	0.5016	-	-	0.5733	0.5667
	UniMEEC [55]	0.4988	0.5929	0.5461	-	-	0.6367	0.5918
	HiLo [56]	0.5178	0.5969	0.5545	0.3304	0.3581	-	0.7197
	M ³ HG [57]	-	-	-	0.4007	0.4196	-	-
	AHMRC [58]	0.5928	<u>0.5600</u>	0.5760	-	-	-	<u>0.7270</u>
Generative Models	GMEC [64]	-	-	<u>0.5945</u>	0.4152	<u>0.4430</u>	-	0.7012
	Video-Llama2-7B	0.5992	0.5301	<u>0.5626</u>	0.4087	<u>0.4201</u>	<u>0.6396</u>	0.7123
	LLaVA1.5-7B	0.5963	0.5088	0.5491	0.4081	0.4241	0.6201	0.6986
	Qwen2.5-Omni-7B	0.5950	0.5397	0.5660	<u>0.4185</u>	0.4272	0.6392	0.7243
Ours	DEC-Debate	0.6951	0.5552	0.6173	0.4563	0.4687	0.6616	0.7620

a seven-class classification task that identifies the emotional classification of an utterance; CE considers whether candidate causal utterances in conversation are correctly extracted with pairs. Finally, Fig. 6 shows the distribution of results for emotion recognition after processing with the DEC-Debate method. It can be observed that after the debate process, the emotion classification distribution shifts toward the diagonal (that is, correct emotion classifications), suggesting that the emotion boundaries were refined.

TABLE IV

EMOTION RECOGNITION ON THE MELD AND EMOTION-CAUSE PAIR EXTRACTION ON THE CONVECP. * DENOTES RESULTS CITED FROM HU ET AL. [55] AND # DENOTES RESULTS CITED FROM JIAN ET AL. [65].

Methods	MELD w-avg.F1	ConvECPE		
		P	R	F1
MMGCN* [61]	0.5865	0.4218	0.4267	0.4211
MM-DFN* [62]	0.5865	<u>0.4679</u>	<u>0.5036</u>	<u>0.4850</u>
UniMSE* [59]	0.6551	0.4424	0.4933	0.4669
GA2MIF* [60]	0.5894	0.4654	0.4859	0.4740
DualGATs# [66]	0.6696	-	-	-
EmoTrans# [67]	0.6739	-	-	-
EmoKSD# [65]	<u>0.6806</u>	-	-	-
DEC-Debate	0.6818	0.5018	0.5054	0.5036

E. Generalization

DEC-Debate framework is validated on the MELD and ConVECP datasets (Table IV). The MELD dataset has received considerable attention in the Emotion Recognition task, so several currently advanced methods are included for comparison. DualGATs [66] is designed to capture the structural relationships between utterances and the emotional dependencies between speakers. EmoTrans [67] highlights the importance and effectiveness of utilizing emotion transition information in emotion recognition tasks. EmoKSD [65] enhances the model’s ability to recognize subtle emotions in conversations.

The experimental results indicate that the ability to extract emotion-cause pairs is not necessarily positively correlated

with the performance of emotion recognition. A model that identifies more emotional utterances will expand the set of candidate emotion-cause pairs, thus increasing the total number of predicted pairs. However, if the model lacks sufficient capability to extract genuine emotion-cause pairs, its precision will decrease, which in turn lowers the F1 score. Likewise, DEC-Debate can identify more emotional utterances and thus yield more candidate emotion-cause pairs, but it leverages debate to reason about the causes of emotions, thereby enabling it to accurately identify true emotion-cause pairs.

F. Ablation Study

The results shown in TABLE V demonstrate the effectiveness of each module in the DEC-Debate. The emotion name (cate) entry denotes the $F1^{cate}$ score for the extraction of emotion-cause pairs dominated by the corresponding emotion.

EBA improves the weight score of pairing by correcting utterances’ emotion classifications, exerting a notable impact on the extraction of emotion-cause pairs for “disgust” and “fear”, and its removal results in a 13.76% decrease in ER scores. Furthermore, EBA can boost the sum of weighted F1 scores for emotion-cause pairs across all emotion categories by refining the emotion classification of utterances. Meanwhile, ECPRA improves emotion-cause pair extraction scores by retaining the correct pairs; its removal results in decreases of 3.38% and 4.00% in pair extraction and CE, respectively. Since ECPRA only retains the emotion-cause pairs validated through debate, its introduction results in a reduction in the number of predictions, with a small number of positive pairs being discarded, leading to a slight decrease in recall. However, the retained emotion-cause pairs have a higher confidence level, indicating an increase in the proportion of positive pairs in the predicted pairings, which boosts precision and ultimately elevates the F1 score. Finally, introducing conversation samples enhanced by Modality Translation (MT) into the debate enables debaters to acquire more contextual information and identify more clues, thereby retaining more emotion-cause pairs and increasing the number of both positive and negative

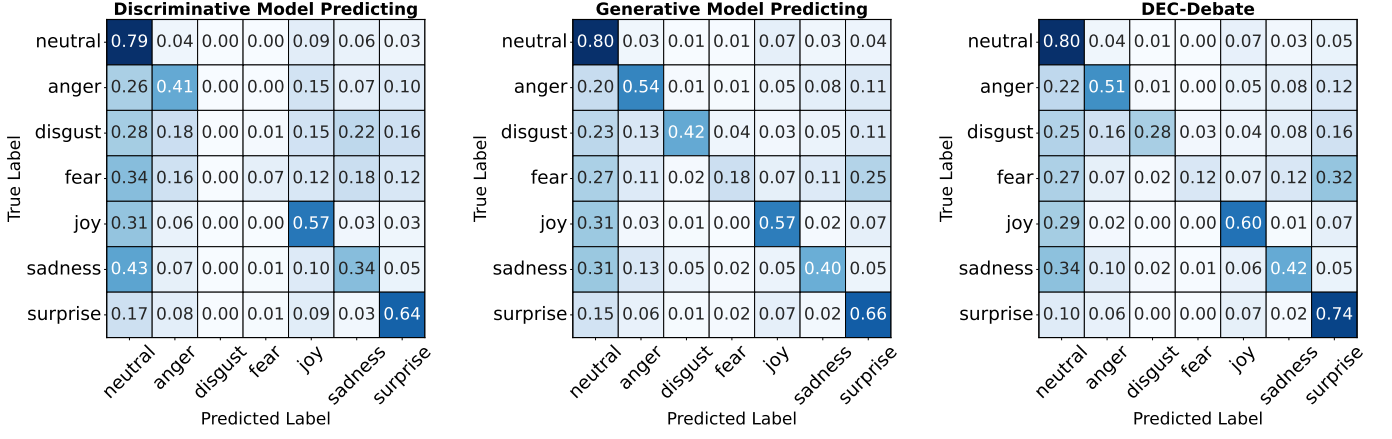


Fig. 6. Row-normalized confusion matrix of emotion recognition results. Each element represents the proportion of utterances with a given true label that are predicted as a specific label.

TABLE V
ABLATION STUDY RESULTS SHOWING THE EFFECT OF EBA, ECPRA AND MODALITY TRANSLATION (MT).

Methods	Anger	Disgust	Fear	Joy	Emotion-Cause Pair Extraction				F1	w-avg.6	w-avg.4	ER w-avg.F1	CE F1
					Sadness	Surprise	P	R					
DEC-Debate	0.4129	<u>0.3648</u>	<u>0.2187</u>	0.5444	0.3510	0.5313	<u>0.6951</u>	<u>0.5552</u>	0.6173	0.4563	0.4687	0.6616	0.7620
w/o EBA	0.3197	0.0000	0.0000	0.4812	0.2597	0.4971	<u>0.6951</u>	<u>0.5552</u>	0.6173	0.3658	0.3986	0.5705	0.7620
w/o ECPRA	<u>0.4057</u>	0.3977	0.2816	0.5018	0.3493	0.4791	0.5247	0.6908	0.5964	0.4347	0.4414	0.6616	0.7315
w/o MT	0.4046	0.3521	0.1999	<u>0.5210</u>	0.3260	<u>0.5236</u>	0.7640	0.4874	0.5951	<u>0.4406</u>	<u>0.4528</u>	<u>0.6458</u>	<u>0.7455</u>

pairs. This results in a slight decrease in precision but an improvement in recall.

TABLE VI
THE IMPACT OF DIFFERENT PROCESSING METHODS ON THE PERFORMANCE OF DUAL MODEL PREDICTION DISCREPANCIES.

Methods	Emotion-Cause Pair Extraction		
	P	R	F1
Discriminative Model	0.6308	0.5045	0.5606
Generative Model	0.5753	0.5504	0.5626
common	0.8903	0.3641	0.5168
discre ^D	0.3969	0.1655	0.2336
discre ^G	0.3644	0.1772	0.2385
discre ^{correct}	1.0000	0.3267	0.4925
discre ^{Voting}	0.3758	0.1542	0.2187
discre ^{DEC-Debate}	0.4904	0.1911	0.2750
common + discre ^{all}	0.5247	0.6908	0.5964
common + discre ^{correct}	0.9390	0.6908	0.7960
Majority Voting	0.6225	0.4978	0.5532
DEC-Debate	<u>0.6951</u>	<u>0.5552</u>	<u>0.6173</u>

To more intuitively demonstrate the effectiveness of the debate mechanism in leveraging dual-model prediction discrepancies to extract performance gain, we further designed and compared multiple discrepancy processing strategies (TABLE VI). The discriminative model achieves higher precision, while the generative model yields higher recall. This indicates that the discriminative model tends to make conservative predictions relying on explicit semantic clues, thus producing fewer but more accurate results; in contrast, the generative model can capture more potential emotion-cause pairs, yet suffers from relatively low precision due to over-inference.

These results confirm that the two models each possess unique strengths, which give rise to discrepancies and endow such discrepancies with exploitable potential. Additionally, the "common" subset refers to consistent predictions from both models; the high precision suggests that consistent predictions have sufficient confidence and can be directly included in the final output. The low recall of this subset stems not from methodological flaws but from its small proportion in the overall test set. For the discrepancy set, we investigate the following strategies, respectively. (1) discre^D: retain only the predictions of the discriminative model; (2) discre^G: retain only the predictions of the generative model; (3) discre^{correct}: retain only the correct predictions within it; (4) discre^{Voting}: determine predictions via a no-debate majority voting mechanism; (5) discre^{DEC-Debate}: determine predictions through the debate. The recall of the discre^{correct} is significantly improved, indicating that the discrepancy set contains mineable correct predictions with considerable value. Ideally, "common + discre^{correct}" constitutes the theoretical performance upper bound of the strategy based on dual model prediction discrepancies, which can be used to evaluate the proximity and optimization space of the current methods. Retaining only the predictions from a single model would lead to the loss of such potential gains. And, reliance solely on majority voting results in a decline in overall performance. In contrast, the introduction of the debate improves both precision and recall, verifying its ability to extract more correct predictions.

The impact of debate rounds n and the number of LLM debaters m on the performance of emotion-cause pair extraction has also been experimentally verified (Fig. 7). The

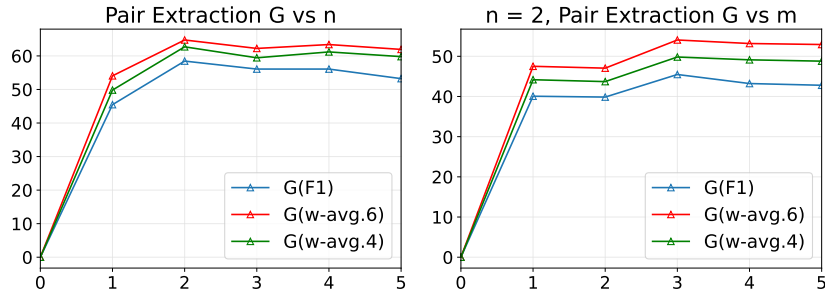


Fig. 7. Results on the ECF dev set. The debate round n and the number of LLMs participating in the debate m have a significant impact on the performance gain of pair extraction. $G(V) = \frac{\Delta V}{V} \times 100\%$.

TABLE VII
THE IMPACT OF INTRODUCING MULTIMODAL INFORMATION ON PERFORMANCE. ONLY TEXTUAL MODALITY IS USED IN THE BASELINE EXPERIMENTS.

Methods			Emotion-Cause Pair Extraction										
	+A	+V	Anger	Disgust	Fear	Joy	Sadness	Surprise	P	R	F1	w-avg.6	w-avg.4
Discriminative Model (BERT-based)	$\times$	$\times$	0.3157	0.0854	0.0624	0.3790	0.3479	0.3712	0.6533	0.4436	0.5284	0.3311	0.3537
	$\checkmark$	$\times$	0.3505	0.0695	0.0377	0.4265	0.3411	0.4046	0.5886	0.5355	0.5608	0.3576	0.3844
	$\times$	$\checkmark$	0.3373	0.0826	0.0597	0.4310	0.3085	0.3918	0.6053	0.5216	0.5603	0.3485	0.3730
	$\checkmark$	$\checkmark$	0.3337	0.0000	0.1111	0.4459	0.2821	0.4186	0.6308	0.5045	0.5606	0.3496	0.3778
Generative Model (Qwen2.5-7B-based)	$\times$	$\times$	0.3431	0.2597	0.2133	0.4965	<u>0.3579</u>	0.4871	0.6012	0.4991	0.5455	0.4108	0.4257
	$\checkmark$	$\times$	0.3636	0.3005	0.1739	0.4669	0.3977	0.4789	0.6128	0.5392	0.5737	0.4141	0.4279
	$\times$	$\checkmark$	0.3717	<u>0.3663</u>	0.1505	0.4609	0.3547	0.4403	0.5429	0.5333	0.5381	0.4021	0.4115
	$\checkmark$	$\checkmark$	0.3612	0.3850	0.2439	0.4785	0.3478	0.4929	0.5753	0.5504	0.5626	0.4178	0.4248
DEC-Debate	$\times$	$\times$	0.4046	0.3521	0.1999	0.5210	0.3260	0.5236	0.7640	0.4874	0.5951	0.4406	0.4528
	$\checkmark$	$\times$	0.4069	0.3586	<u>0.2295</u>	0.5287	0.3368	<u>0.5271</u>	<u>0.7443</u>	0.5098	0.6051	0.4470	0.4587
	$\times$	$\checkmark$	0.4087	0.3586	0.2258	<u>0.5359</u>	0.3381	0.5259	0.7265	0.5221	<u>0.6076</u>	<u>0.4494</u>	<u>0.4613</u>
	$\checkmark$	$\checkmark$	0.4192	0.3648	0.2187	0.5444	0.3510	0.5313	0.6951	0.5552	0.6173	0.4563	0.4687

experimental results show that when the number of debate rounds $n \geq 2$, system performance stabilizes, indicating that the process has approached equilibrium, i.e., a state in which no significant increase in benefits can be achieved even if more rounds of debate are conducted, and that a final decision can reasonably be drawn at this point. The maximum performance gain is achieved when $m = 3$. Therefore, we choose $n = 2, m = 3$ to achieve the best results.

TABLE VIII
THE IMPACT OF DIFFERENT MULTIMODAL FUSION METHODS APPLIED ON THE RESULTS OF DEC-DEBATE. ITALICS INDICATE CONSISTENCY WITH THE MAIN EXPERIMENTAL SETUP.

	P	R	F1	P	R	F1
Fusion Methods in Discriminative Model						
+Add	0.3352	0.2808	0.3056	0.6405	0.3053	0.4135
+Concat	0.3120	0.2637	0.2858	0.6298	0.2816	0.3935
+GCN	0.4513	0.3048	0.3639	0.6776	0.3299	0.4438
+MH-Att	<u>0.6056</u>	0.4468	0.5142	<u>0.7413</u>	0.4255	0.5407
+ASF [44]	0.6308	0.5045	0.5606	<i>0.6951</i>	0.5552	0.6173
Fusion Methods in Generative Model						
+Omni	0.5950	<u>0.5397</u>	0.5660	0.7499	<u>0.4933</u>	<u>0.5951</u>
+MT	0.5753	0.5504	<u>0.5626</u>	<i>0.6951</i>	0.5552	0.6173

Results of experiments on modality ablation (TABLE VII) indicate that acoustic information is more effective during the fine-tuning of discriminative and generative models, while visual information may carry more interference. In contrast, visual information can assist debaters in identifying con-

versational contexts during debate, and thus exerts a more significant effect. Notably, for the modal ablation experiments of DEC-Debate, all modalities are incorporated into the base prediction models. TABLE VIII shows that simple fusion methods, such as addition and concatenation, yield poor performance. ASF achieves complementary fusion by adding a modal transformation on the basis of multi-head attention (MH-Att), and thus is more effective. At present, using the full-modal version (Omni) of LLM with the incorporation of video and audio information greatly increases the difficulty of fine-tuning, which is less convenient and effective than the adoption of Modality Translation (MT).

G. Case Study

Process of Debate Fig. 8 demonstrates how the debate mechanism reasons. In EBA, Debater1 and Debater2 initially believed that Rachel’s dissatisfaction with the appearance of clothes reflected the emotion of disgust. In contrast, Debater3 argued that Rachel’s words sounded more like a mild complaint coupled with resignation, without expressing intense emotion, and thus leaned toward classifying the emotion as sadness. In the second round of debate, Debater1 was persuaded by Debater 3 and acknowledged that labeling the emotion as “Disgust” was too intense, subsequently revising its judgment. Ultimately, the emotion category of utterance 2 was correctly identified as “Sadness,” which matches the actual label for this sample. The process reveals that the LLM debater is capable of correcting its own decision after reviewing a more

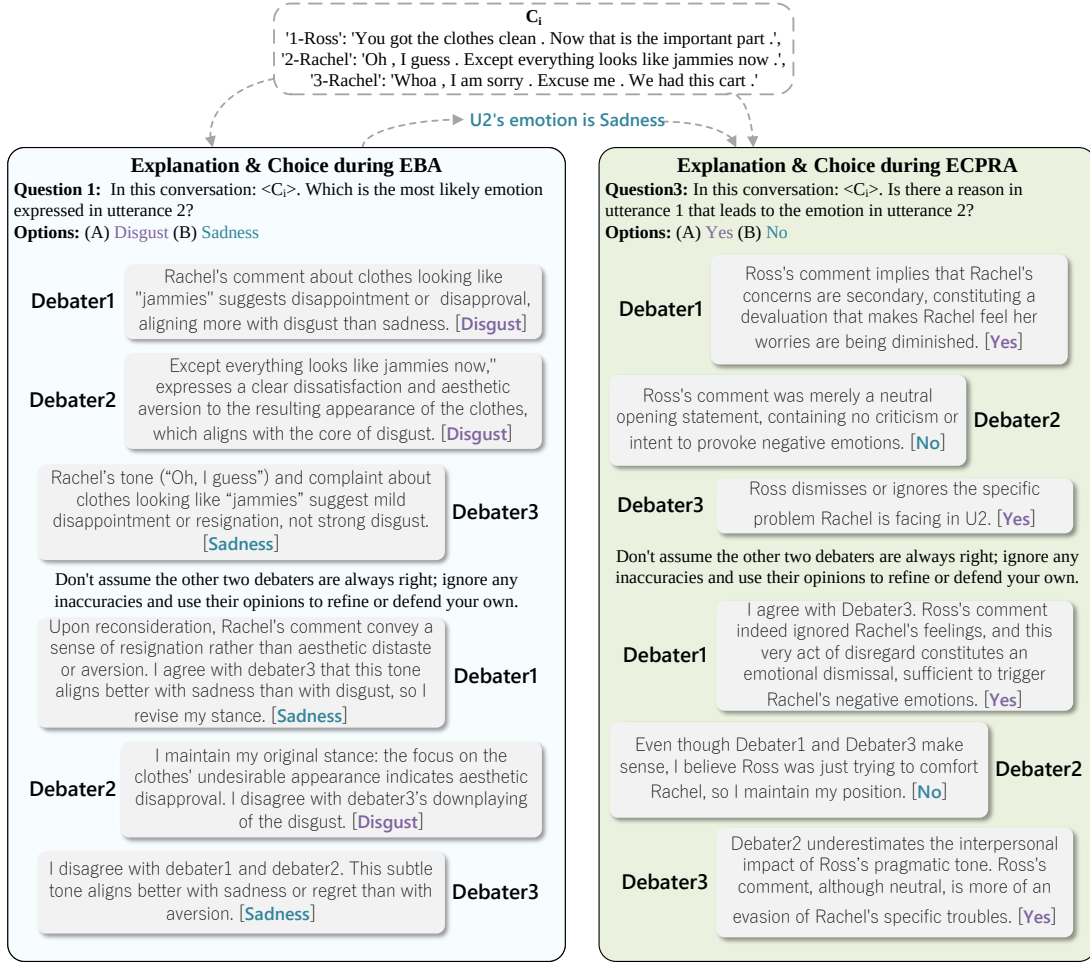


Fig. 8. A brief demonstration of the debate process. The generative model predicts U2's emotion as Disgust and considers U1 to be the corresponding causal utterance; whereas the discriminative model predicts U2's emotion as Sadness and does not consider U1 to be the corresponding causal utterance.

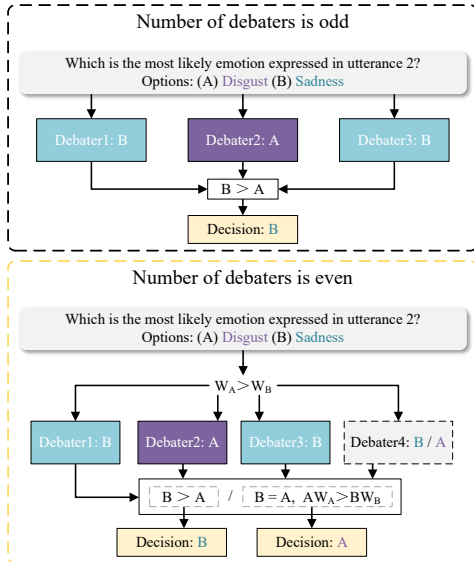


Fig. 9. The Decision-Making mechanism of debate.

rational explanation. In ECPRA, Debater1 and Debater3 both considered Ross's comment in utterance 1 as one of the causes

that triggered the emotional response of Rachel in utterance 2, arguing that Ross failed to take Rachel's feelings into account. Debater2, on the other hand, viewed the comment of Ross as neutral, containing no criticism or blame, and therefore should not be regarded as a cause of Rachel's sadness. In the second round, Debater1 and Debater3 both maintained their original judgments, asserting that Debater2 failed to fully grasp the negative implications of Ross's ostensibly neutral comment within the specific context. Nevertheless, Debater2 did not yield despite facing opposition from both Debater1 and Debater3. Instead, Debater2 introduced a new explanation. Although Debater2 ultimately adhered to an incorrect choice, it somewhat demonstrates that the LLM debater is capable of providing justification for what it believes to be the correct choice, rather than readily accepting other viewpoints. Throughout the debate process, each debater was able to access and respond to the viewpoints of others, engaging in mutual rebuttals. This multi-round interaction prompted the models to conduct deeper and more structured reasoning by comprehensively evaluating diverse perspectives, thereby increasing the confidence of the final prediction. By simulating human-like reasoning in complex scenarios, this mechanism effectively enhances the ability of the system to capture the

				
Label	Anger $\Leftarrow$ U1	Joy $\Leftarrow$ U2	Neutral	Neutral
Generative	Anger $\Leftarrow$ U1	Joy $\Leftarrow$ U2, U1	Fear $\Leftarrow$ U3	Neutral
Discriminative	Anger $\Leftarrow$ None	Joy $\Leftarrow$ U2	Neutral	Neutral
Discrepancies	Anger $\Leftarrow$ U1 or None?	Joy $\Leftarrow$ U1 ?	Fear or Neutral? $\Leftarrow$ U3 ?	None
EBA			Fear	
ECPra	Anger $\Leftarrow$ U1	Joy $\Leftarrow$ U1	Fear $\Leftarrow$ U3	
Final Decision	Anger $\Leftarrow$ U1	Joy $\Leftarrow$ U2	Fear $\Leftarrow$ None	Neutral

Fig. 10. In the DEC-Debate framework, the discrepancies generated by different base predictions are addressed by EBA and ECPRA for the confirmation of emotion classification and emotional cause positions, respectively.

relationship between emotions and their underlying causes.

Decision-Making Mechanism Each debate involves exactly two candidate options, one provided by a discriminative model and the other by a generative model, and the issue of deadlock can be effectively avoided through the following mechanism (Fig. 9): (1) When the total number of debaters m is odd, any option receiving support exceeding more than half of the debaters (i.e., $> \frac{m}{2}$) is selected as the final outcome. (2) When m is even, a scenario may occur where the two predictions are supported by an equal number of debaters. To address this, different weights can be assigned to these predictions by comparing the performance of the base prediction models. Given that the discriminative model exhibits slightly weaker performance than the generative model in our experiments, its proposed prediction must receive strictly more than half of the votes ($> \frac{m}{2}$) to be adopted. In contrast, a prediction from the generative model requires only at least half the votes ($\geq \frac{m}{2}$) to be adopted as the final result.

Discrepancies in Debate The conversation illustrated in Fig. 10 depicts a scenario where Chandler is trapped with others, among whom U1, U2, and U3 are all Chandler's inner monologues. The "anger" expressed in U1 is identified by both the generative model and the discriminative model, yet the two base models diverge in their predictions regarding whether there exists a trigger for this emotion in U1. Similarly, discrepancies arise between the two models when predicting whether the U1 contains the cause of the emotion in U2. In the analysis of U3, the generative model contends that U3 conveys the emotion of "fear", while the discriminative model categorizes it as "neutral". This divergence further extends to the prediction of the cause of the emotion in U3. These discrepancies mentioned above arise from the distinct prediction paradigms of the two types of models elaborated earlier in this paper: the generative model adopts a more open prediction approach, while the discriminative model's predictions rely more heavily on explicit cues. After processing by the debate mechanism, the over-predictions of the generative

model (i.e., the assertions that the cause of U2's emotion lies in U1 and that U3 contains the cause of its own emotion) are suppressed. Meanwhile, the "U1-U1" emotion-cause pair that the discriminative model fails to identify is incorporated into the final result, which improves the accuracy of the outcome.

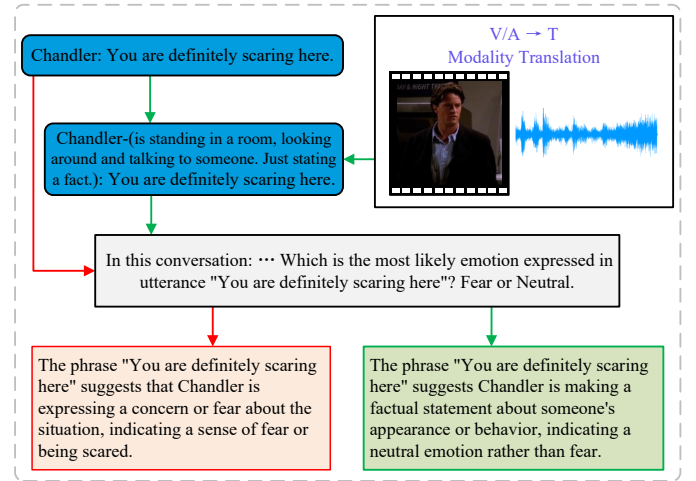


Fig. 11. In debate, using original conversation texts and conversation texts enhanced by Modality Translation can yield different results.

Effectiveness of Modality Translation The case illustrated in Fig. 10 presents a debate result based solely on textual information, where EBA failed to correctly revise the emotion of U3 to "neutral". This limitation arises because the model lacks access to visual and acoustic information. However, after enriching the conversation text with visual and acoustic descriptions via Modality Translation (Fig. 11), debaters could infer that Chandler exhibited no abnormal behavior and spoke in a factual tone, thereby correctly revising the emotion of U3 to "neutral" rather than "fear".

H. Efficiency Analysis

The debate mechanism introduces an additional inference overhead. We experiment with and measure the per-conversation average inference latency introduced by debate and compare it with baseline methods that also adopt generative LLMs. The results (TABLE IX) demonstrate that the debate mechanism achieves a favorable trade-off between efficiency and performance. This stems from two factors: (1) the debate mechanism effectively harnesses multiple quantized LLMs, and (2) it targets prediction discrepancies rather than full re-prediction. Compared to methods with similar or larger-scale models, debate achieves superior performance with faster inference. In contrast, Question-Answer (QA) and Chain-of-Thought (CoT) are designed to completely infer conversations and struggle to improve performance. QA relies heavily on strong LLMs, while CoT suffers from prohibitively long inference times (T/O) on large models due to its long thought chains in this task. The method that enhances predictions via LLMs further exacerbates latency by requiring extensive secondary inference over numerous candidate predictions.

TABLE IX
AVERAGE INFERENCE COST IN EMOTION-CAUSE PAIR EXTRACTION.

Methods	Models	Scale	Latency(ms)	F1
QA [56]	GPT	>175B	13,672	0.2026
	Qwen2.5	32B	2,517	0.0744
CoT [38]	FlanT5	0.25B	9,660	0.1326
	Qwen2.5	32B	T/O	-
enhance [64]	GPT	>175B	33,103	<u>0.5945</u>
	Qwen2.5	32B	26,927	0.5697
debate	Qwen2.5	7B	<u>5,398</u>	0.6173
	GLM4	9B		
	Llama3.1	8B		

I. Error Analysis

As shown in Fig. 10, EBA failed to correctly identify the emotion of U3 as "neutral." Although the model made the correct prediction after multimodal descriptions were introduced (Fig. 11), this reveals a limitation of DEC-Debate in understanding subtle and artistically nuanced conversation scenarios. Specifically, while EBA accurately labeled U3's emotion as "neutral" after incorporating multimodal information, the debaters did not recognize that this utterance is actually Chandler's internal monologue. Instead, they inferred based on Chandler's external actions and tone that U3 represents a statement rather than an expression of emotion. Although this led to the correct outcome, their underlying scene modeling was flawed. In fact, even human observers would find it challenging to immediately grasp Chandler's behavioral intent in this context without sufficient familiarity with the narrative background. This suggests that, at its current stage, DEC-Debate performs well in conventional conversations but still lacks sufficient capability to interpret conversation scenes that are imbued with artistic.

Furthermore, sarcasm, a common linguistic phenomenon that often follows specific emotions, is related to emotion-cause analysis. In some instances, debaters can perceive the sarcastic tone in utterances, which facilitates the inference of emotions and their causes. However, sarcasm relies on tension between literal meaning and real intent. Its high semantic ambiguity may still mislead debaters, resulting in emotional misjudgments or forming wrong emotion-cause relationships.

V. CONCLUSION

This paper introduces a suitable debate framework (DEC-Debate) for the MECPEC task, aimed at extracting more reasonable emotion-cause pairs from predictions given by different models. DEC-Debate consists of two main modules: EBA and ECPRA. EBA and ECPRA encourage LLMs to argue with each other, picking better emotion classifications and retaining emotion-cause pairs with clear clues. They leverage the prediction discrepancies between a discriminative model and a generative model to improve the performance of the task. Advanced performance has been achieved in the MECPEC tasks, demonstrating the effectiveness of DEC-Debate.

Human utterances can be implicit or ambiguous, posing challenges to the practical application of affective HCI systems. The method we propose can improve affective HCI in the following ways. (1) DEC-Debate enhances the model's ability to understand the causes behind emotions, which can be leveraged to enable systems to gain deeper insights into the underlying emotions and their reasons in conversations, thereby generating responses that are better aligned with human needs. (2) Integration of debate mechanisms into various specific tasks essentially constitutes an exploration of debate as a rich and vivid form of natural language, which may open up new avenues to facilitate the naturalness of interactions in AI-based HCI systems. (3) Prior to the output of final results, the method refutes flawed reasoning through debate, mitigating the impact of erroneous information; meanwhile, it generates an interpretable natural language reasoning process, which can be used to enhance the credibility of HCI systems.

VI. FUTURE WORK

Although this study validates the effectiveness of the DEC-Debate, it recognizes certain limitations that point to potential avenues for future work. First, the choices provided by the two base models are not always appropriate. It is worth exploring how to introduce more options during the debate. In the next step, we will explore the possibility of allowing the debaters to provide additional answers they believe to be correct if they determine that the initially provided predictions do not contain the right one. This change may have a significant impact on the EBA. Secondly, incorporating stronger multimodal LLMs into the debate process will also be considered. With advances in technology, the multimodal perception capabilities of LLMs are rapidly improving. Including richer modal information in the debate may lead to more comprehensive and thorough discussions. For example, debaters could establish connections across different modalities, enabling faster identification of emotion causes and reducing the reliance on understanding

the conversational context. It is expected to have a greater positive impact on ECPRA. Notably, this study focuses more on designing and validating an overall framework and process suitable for affective computing, rather than concentrating on training a complete model. Therefore, it can be relatively easily adapted to other tasks. In the future, we plan to apply this framework to more complex tasks, such as generating emotional causes, which can further leverage the generation capabilities of the LLMs involved in the debate.

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